

Lessons 3-4: Reentry EOM Development

MECH 637: Astrodynamic Reentry

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Equations of Motion

- Six independent quantities are required to define the motion of a point mass at any given time, with these quantities being three components each of position and velocity.
- For reentry, the equations of motion (EOM) will use three position quantities, a velocity magnitude, and two angles describing the direction of the velocity.
- In terms of Newton's Second Law and with respect to an inertial reference frame, the total force acting on a point mass at any instant in time is given by the following:

Where mass (m) is assumed constant, $\vec{T}$ is the thrust force, $\vec{A}$ is the aerodynamic force, and $\vec{g}$ is the gravitational force. Overall, the notation " I " superscript emphasizes that the derivative is with respect to an inertial reference frame.

- For reentry, however, motion is more conveniently measured relative to a planetary-fixed, or rotating, reference frame rather than inertial. Recalling from dynamics, the inertial derivatives for motion can be expressed in a rotating reference frame by:

Where the "R" superscript denotes a derivative in a rotating reference frame, and $\vec{\omega}$ is the general angular velocity of the rotating frame relative to the inertial frame.

- Assuming that the preceding rotating reference frame is same as the planet-fixed $OX_1Y_1Z_1$ system, then the inertial derivative for acceleration can simplified since $\vec{\omega} = \vec{\omega}_{\oplus}$ is a constant vector:

Note 1: The distributive property for cross products is defined as:

$$\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$$

Note 2: For cross products, the product rule of differential calculus is defined as:

$$\frac{d}{dt}(\vec{a} \times \vec{b}) = \frac{d\vec{a}}{dt} \times \vec{b} + \vec{a} \times \frac{d\vec{b}}{dt}$$

Since $\vec{\omega}_{\oplus}$ is a constant vector, then:

Note 3: The distributive property for cross products with scalar multiplication is:

$$(r\vec{a}) \times \vec{b} = \vec{a} \times (r\vec{b}) = r(\vec{a} \times \vec{b})$$

$$\underbrace{\quad}_1 \quad \underbrace{\quad}_2 \quad \underbrace{\quad}_4$$

$\underbrace{\quad}_3$

1. Relative acceleration as seen from the rotating reference frame
 2. Relative velocity as seen from the rotating reference frame
 3. Coriolis acceleration
 4. Centripetal acceleration
- Introducing Newton's Second Law gives the following:
 - For convenience, the term can be introduced as the velocity relative to the planet:

Kinematic Equations

- In the vehicle-pointing $OX_2Y_2Z_2$ system, the position vector $\vec{r}$ can be expressed as:

$$\vec{r} = r\hat{e}_{x_2}$$

- In the velocity-referenced $OX''Y''Z''$ system, the relative velocity vector is expressed as:
- Using the coordinate transformation relating the vehicle-pointing and velocity referenced systems, then the equation for becomes:

$$\begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \cos\gamma & -\sin\gamma\cos\psi & -\sin\gamma\sin\psi \\ \sin\gamma & \cos\gamma\cos\psi & \cos\gamma\sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix}$$

Note: The preceding equation represents the velocity as viewed (or measured) by an observer on the rotating planet-fixed system, but written in terms of the vehicle-pointing system.

- Recalling from dynamics, the derivative of a vector in one reference frame is related to the derivative in another frame by the following:

Where the superscripts "1" and "2" denote the reference frame, and the subscript "2/1" indicates the angular rate of reference frame 2 relative to frame 1.

- With representing the velocity with respect to the planet-fixed $OX_1Y_1Z_1$ system in terms of the vehicle-pointing $OX_2Y_2Z_2$ system, then the following can be developed:

$$= \dot{r}\hat{e}_{x_2} + (r\dot{\theta}\cos\phi)\hat{e}_{y_2} + (r\dot{\phi})\hat{e}_{z_2}$$

- Combining the two equations for in terms of the vehicle-pointing $OX_2Y_2Z_2$ system yield the kinematic equations. When integrated, these equations provide the position of the entry vehicle as observed from the rotating planet.

Force Equations

- The kinematic equations are not a complete set of equations of motion, because they only provide three independent quantities r , θ , ϕ , rather than the full six required to define the motion of a point mass at any given time. Additional equations will be derived through an examination of the right-hand side of the relative acceleration equation:

- Previously, the rotation vector of a planet-fixed system with respect to the inertial system was defined as the following equation, expressed in vehicle-pointing coordinates:

$$\vec{\omega}_{\oplus} = [\vec{\omega}]_{1/0}^{\hat{e}_2} = (\omega_{\oplus} \sin \phi) \hat{e}_{x_2} + (\omega_{\oplus} \cos \phi) \hat{e}_{z_2}$$

Using this equation, the operation becomes:

With $\vec{r} = r \hat{e}_{x_2}$, then the operation $\vec{\omega}_{\oplus} \times \vec{r}$ yields:

$$\vec{\omega}_{\oplus} \times \vec{r} = \begin{vmatrix} \hat{e}_{x_2} & \hat{e}_{y_2} & \hat{e}_{z_2} \\ \omega_{\oplus} \sin \phi & 0 & \omega_{\oplus} \cos \phi \\ r & 0 & 0 \end{vmatrix} = (r \omega_{\oplus} \cos \phi) \hat{e}_{y_2}$$

Next, evaluating the centripetal acceleration term gives:

$$\begin{aligned} \vec{\omega}_{\oplus} \times (\vec{\omega}_{\oplus} \times \vec{r}) &= \begin{vmatrix} \hat{e}_{x_2} & \hat{e}_{y_2} & \hat{e}_{z_2} \\ \omega_{\oplus} \sin \phi & 0 & \omega_{\oplus} \cos \phi \\ 0 & r \omega_{\oplus} \cos \phi & 0 \end{vmatrix} \\ &= - (r \omega_{\oplus}^2 \cos^2 \phi) \hat{e}_{x_2} + (r \omega_{\oplus}^2 \sin \phi \cos \phi) \hat{e}_{z_2} \end{aligned}$$

- Recalling that $\vec{F} = \vec{T} + \vec{A} + m\vec{g}$, now each force term needs to be expressed in the vehicle-pointing system. Starting with right-side, the gravitational force ($\vec{g}$) always acts along the radius vector:

$$m\vec{g} = -mg(r) \hat{e}_{x_2}$$

- The aerodynamic force ($\vec{A}$) can be broken into drag and lift components. With the former, the drag force ($\vec{D}$) acts in a direction opposite to the velocity equation. In the velocity-referenced $OX''Y''Z''$ system, the drag vector is expressed as:

$$\vec{D} = -D \hat{e}_y'' = -D \hat{e}_v$$

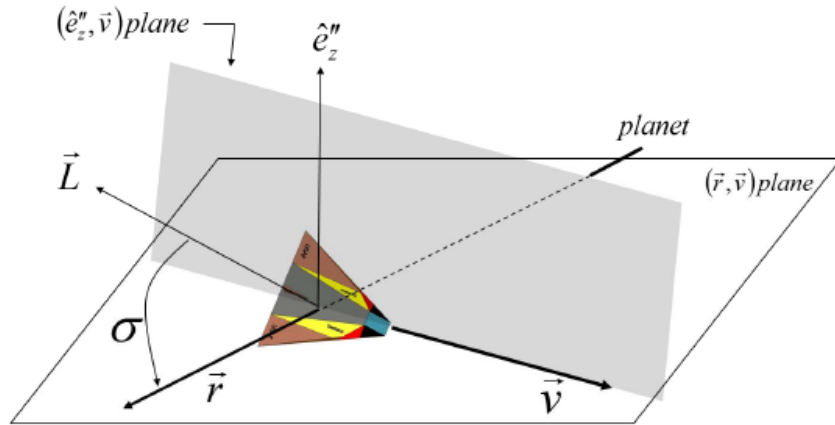
- Using the coordinate transformation relating the vehicle-pointing $OX_2Y_2Z_2$ and velocity referenced $OX''Y''Z''$ systems, then the equation for $\vec{D}$ becomes:

$$\begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \cos\gamma & -\sin\gamma \cos\psi & -\sin\gamma \sin\psi \\ \sin\gamma & \cos\gamma \cos\psi & \cos\gamma \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ -D\hat{e}_y'' \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\gamma & -\sin\gamma \cos\psi & -\sin\gamma \sin\psi \\ \sin\gamma & \cos\gamma \cos\psi & \cos\gamma \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix}$$

$$\vec{D} = -D\hat{e}_y'' = -(D\sin\gamma)\hat{e}_{x_2} - (D\cos\gamma\cos\psi)\hat{e}_{y_2} - (D\cos\gamma\sin\psi)\hat{e}_{z_2}$$

- The lift force ($\vec{L}$) acts perpendicular to the velocity vector, and orientation of $\vec{L}$ relative to the $(\vec{r}, \vec{v})$ plane is defined with the bank angle (σ):



- As shown in the preceding figure, the angle between $\vec{L}$ and the $\hat{e}_z''$ -axis is $(90^\circ - \sigma)$. Based on this angular separation, a coordinate transformation can be developed that aligns a “new” z -axis $\hat{e}_z'''$ with $\vec{L}$ to create a $OX_3Y_3Z_3$ reference frame:

$$\begin{bmatrix} \hat{e}_x''' \\ \hat{e}_y''' \\ \hat{e}_z''' \end{bmatrix} = \mathbf{R}_{y''}(90^\circ - \sigma) \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix}$$

$$\begin{bmatrix} \hat{e}_x''' \\ \hat{e}_y''' \\ \hat{e}_z''' \end{bmatrix} = \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \cos(90^\circ - \sigma) & 0 & -\sin(90^\circ - \sigma) \\ 0 & 1 & 0 \\ \sin(90^\circ - \sigma) & 0 & \cos(90^\circ - \sigma) \end{bmatrix} \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix}$$

$$= \begin{bmatrix} \sin\sigma & 0 & -\cos\sigma \\ 0 & 1 & 0 \\ \cos\sigma & 0 & \sin\sigma \end{bmatrix} \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix}$$

- The $OX_3Y_3Z_3$ frame is related to the vehicle-pointing $OX_2Y_2Z_2$ system with the following coordinate transformation:

$$\begin{aligned} \begin{bmatrix} \hat{e}''' \end{bmatrix} &= \mathbf{R}_{y''}(90^\circ - \sigma) \mathbf{R}_{z'}(-\gamma) \mathbf{R}_{x_2}(\psi) \begin{bmatrix} \hat{e}_2 \end{bmatrix} \\ &= \begin{bmatrix} \sin\sigma & 0 & -\cos\sigma \\ 0 & 1 & 0 \\ \cos\sigma & 0 & \sin\sigma \end{bmatrix} \begin{bmatrix} \cos\gamma & -\sin\gamma \cos\psi & -\sin\gamma \sin\psi \\ \sin\gamma & \cos\gamma \cos\psi & \cos\gamma \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix} \\ &= \begin{bmatrix} \hat{c}_1 & \hat{c}_2 & \hat{c}_3 \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix} \end{aligned}$$

Where the unit column vectors are given by:

$$\begin{aligned} \hat{c}_1 &= \begin{bmatrix} \sin\sigma \cos\gamma \\ \sin\gamma \\ \cos\sigma \cos\gamma \end{bmatrix} \\ \hat{c}_2 &= \begin{bmatrix} -\sin\sigma \sin\gamma \cos\psi + \cos\sigma \sin\psi \\ \cos\gamma \cos\psi \\ -\cos\sigma \sin\gamma \cos\psi - \sin\sigma \sin\psi \end{bmatrix} \\ \hat{c}_3 &= \begin{bmatrix} -\sin\sigma \sin\gamma \sin\psi - \cos\sigma \cos\psi \\ \cos\gamma \sin\psi \\ -\cos\sigma \sin\gamma \sin\psi + \sin\sigma \cos\psi \end{bmatrix} \end{aligned}$$

- $\hat{e}_z'''$ -axis: Aligned with the lift vector $\vec{L}$
- $\hat{e}_y'''$ -axis: Aligned with the velocity

- $\hat{e}_x'''$ -axis: Completes the system such that $\hat{e}_x''' \times \hat{e}_y''' = \hat{e}_z'''$; analogous to the pitch axis for aircraft
- Alternatively, the $OX_3Y_3Z_3$ frame can be written as the following set of unit vectors, referred to as the $OX_pY_vZ_L$ reference frame:

$$\begin{bmatrix} \hat{e}_x''' \\ \hat{e}_y''' \\ \hat{e}_z''' \end{bmatrix} = \begin{bmatrix} \hat{e}_p \\ \hat{e}_v \\ \hat{e}_L \end{bmatrix}$$

- In the $OX_pY_vZ_L$ system, the lift vector is expressed as:

$$\vec{L} = L\hat{e}_z''' = L\hat{e}_L$$

- Using the coordinate transformation relating the vehicle-pointing $OX_2Y_2Z_2$ and $OX_pY_vZ_L$ systems, then the equation for $\vec{L}$ becomes:

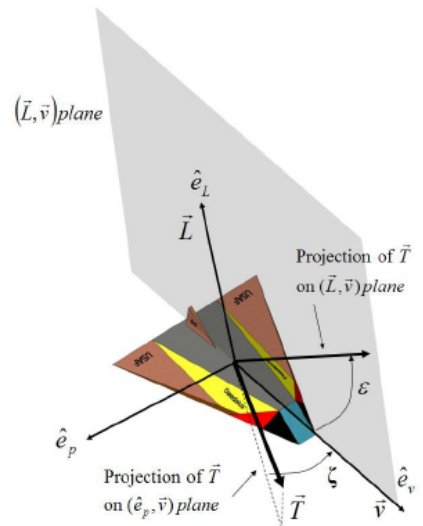
$$\begin{bmatrix} \hat{e}_p \\ \hat{e}_v \\ \hat{e}_L \end{bmatrix} = \begin{bmatrix} \hat{c}_1 & \hat{c}_2 & \hat{c}_3 \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ L\hat{e}_L \end{bmatrix} = \begin{bmatrix} \hat{c}_1 & \hat{c}_2 & \hat{c}_3 \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix}$$

$$\vec{L} = L\hat{e}_L = L(\cos\sigma\cos\gamma)\hat{e}_{x_2} + L(-\cos\sigma\sin\gamma\cos\psi - \sin\sigma\sin\psi)\hat{e}_{y_2} + L(-\cos\sigma\sin\gamma\sin\psi + \sin\sigma\cos\psi)\hat{e}_{z_2}$$

- The final force component, thrust $(\vec{T})$, can be written as the following equation in terms of the $OX_pY_vZ_L$ system for the general case when thrust may not align with the velocity:

$$\begin{aligned} \vec{T} &= T(\sin\zeta)\hat{e}_p + T(\cos\zeta\cos\epsilon)\hat{e}_v + T(\cos\zeta\sin\epsilon)\hat{e}_L \\ &= T \begin{bmatrix} \sin\zeta & \cos\zeta\cos\epsilon & \cos\zeta\sin\epsilon \end{bmatrix} \begin{bmatrix} \hat{e}_p \\ \hat{e}_v \\ \hat{e}_L \end{bmatrix} \end{aligned}$$

The thrust $\vec{T}$ is related to the vehicle-pointing $OX_2Y_2Z_2$ system with the following coordinate transformation:



$$\vec{T} = T \begin{bmatrix} \sin\zeta & \cos\zeta\cos\epsilon & \cos\zeta\sin\epsilon \end{bmatrix} \begin{bmatrix} \hat{c}_1 & \hat{c}_2 & \hat{c}_3 \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix} = \begin{bmatrix} T_1 & T_2 & T_3 \end{bmatrix} \begin{bmatrix} \hat{e}_{x_2} \\ \hat{e}_{y_2} \\ \hat{e}_{z_2} \end{bmatrix}$$

Where the components of thrust are given by:

$$T_1 = T(\sin\zeta\sin\sigma\cos\gamma + \cos\zeta\cos\epsilon\sin\gamma + \cos\zeta\sin\epsilon\cos\sigma\cos\gamma)$$

$$T_2 = T \left[\sin\zeta(-\sin\sigma\sin\gamma\cos\psi + \cos\sigma\sin\psi) + \cos\zeta\cos\epsilon\cos\gamma\cos\psi + \cos\zeta\sin\epsilon(-\cos\sigma\sin\gamma\cos\psi - \sin\sigma\sin\psi) \right]$$

$$T_3 = T \left[\sin\zeta(-\sin\sigma\sin\gamma\sin\psi - \cos\sigma\cos\psi) + \cos\zeta\cos\epsilon\cos\gamma\sin\psi + \cos\zeta\sin\epsilon(-\cos\sigma\sin\gamma\sin\psi + \sin\sigma\cos\psi) \right]$$

Final EOM Development

- When all of the components are assembled, the equation for $\dot{\vec{r}}$ can be expressed in the vehicle-pointing $OX_2Y_2Z_2$ system:
- Similar to the development for $\dot{\vec{r}}$, the derivative of a vector in one reference frame is related to the derivative in another frame by the following:

- In the vehicle-pointing $OX_2Y_2Z_2$ system, the term $\dot{\vec{r}}$ can be computed by differentiating the equation for $\vec{r}$:

- The cross-product term $\vec{\omega} \times \vec{r}$ can be computed as:

- When all of the components are assembled, the equation for $\dot{\vec{r}}$ can be expressed as:

$$\dot{\vec{r}} = a_1\hat{e}_{x_2} + a_2\hat{e}_{y_2} + a_3\hat{e}_{z_2}$$

- After multiplying both sides of the preceding equation by mass (m), a set of three coupled, scalar differential equations can be attained by comparing the following component-by-component:

- Solving the preceding system of differential equations for $\dot{\psi}$ yields:
- In combination, the three kinematic and three force equations give six independent parameters required to define the motion of a point mass at any time.

Summary of Assumptions for EOM Development

- Point mass (for entry vehicle)
- Planetary rotation is constant about the polar axis
- Planet-fixed and inertial reference frames are aligned at $t = 0$
- Constant mass (for entry vehicle)
- Drag force acts in the direction opposite to velocity
- Lift force is perpendicular to velocity
- Gravitational force is directed along a vector from the point mass to the planetary center

Source Consulted

Hicks, Kerry D. *Introduction to Astrodynamic Reentry*, Second Edition. Dayton, OH: Hicks, 2014.